Data Quality Assessment Report

sample_data.xlsx

Sheet: Sheet1

40.0

Overall Quality Score / 100

Poor

Processing Date: 2026-08-05 01:35:14

Engine Version: Phase 1

Execution Time: 0.05s

Generated By: Data Quality Engine (Phase 1)

Executive Summary

Dataset:	sample_data.xlsx
Rows / Columns:	7 rows x 4 columns
Checks Executed:	8
Overall Quality Score:	40.0 / 100 (Poor)

Critical Findings

- Missing Values: 3 column(s) affected, 3 issue(s) found
- Outliers: 1 column(s) affected, 2 issue(s) found
- Consistency: 1 column(s) affected, 1 issue(s) found
- Validity: 1 column(s) affected, 2 issue(s) found
- Freshness: 1 column(s) affected, 1 issue(s) found
- PII detected in 1 column(s) -- masking required before sharing

Positive Findings

- Duplicates: no issues found
- Type Mismatch: no issues found
- Schema Quality: no issues found

Data Readiness

Not Recommended

Business Impact & Risk Summary

Missing Values

Critical

Business Impact: May produce inaccurate analytics and unreliable reporting.

Outliers

Critical

Business Impact: May distort averages, totals, and any ML model trained on this data.

Consistency

High

Business Impact: Grouping and aggregation become unreliable (e.g. 'Paid' vs 'paid' split apart).

Validity

High

Business Impact: Values that look present may still be unusable or logically wrong.

Freshness

High

Business Impact: Decisions may be based on stale data without anyone noticing.

PII Exposure

Critical

Business Impact: Privacy and compliance risk if this data is shared unmasked.

Dataset Overview

Rows:	7
Columns:	4
Header Row:	0
Processing Time:	0.05s

Column Classification

Measurement Columns:	1
Identifier Columns:	1
PII Columns:	1
Free-Text Columns:	0
Categorical Columns:	0
Date Columns:	1

Data Quality Dashboard

Completeness

25

weight 20% | assessed

Validity

0

weight 20% | assessed

Type Reliability

100

weight 15% | assessed

Consistency

0

weight 15% | assessed

Uniqueness

100

weight 10% | assessed

Schema Quality

100

weight 10% | assessed

Outlier Risk

0

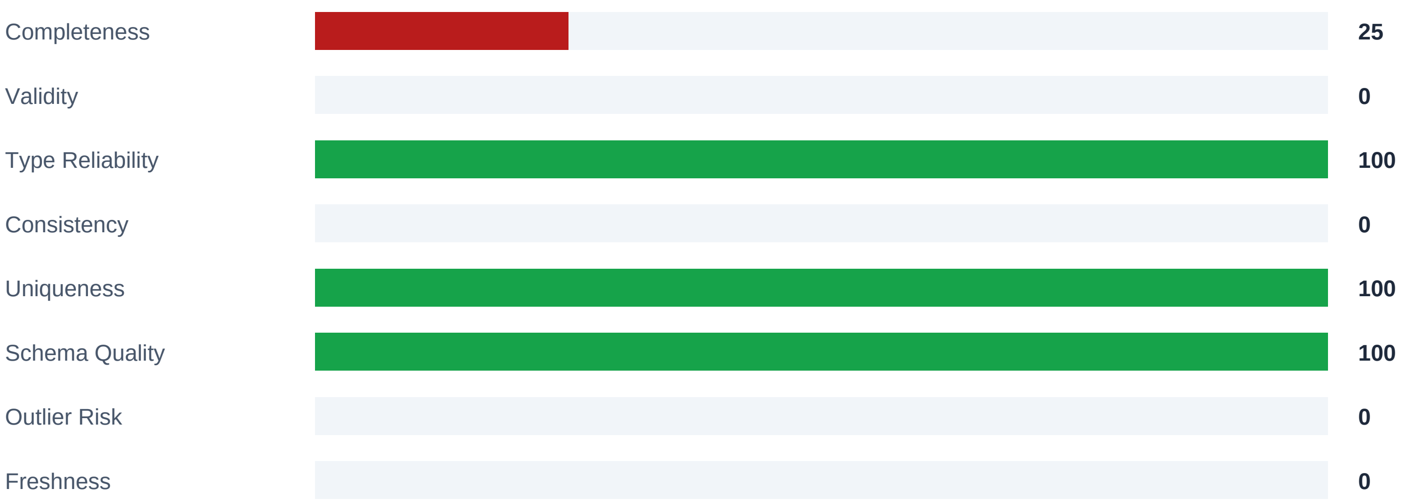
weight 5% | assessed

Freshness

0

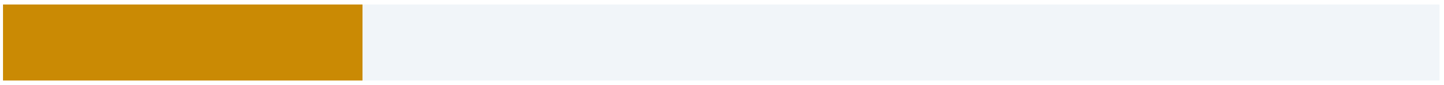
weight 5% | assessed

Dimension Scores at a Glance



Privacy Risk (separate from score above)

MEDIUM



1 column(s) with PII vs. 3 clean column(s) (of 4 total)

Types found: MOBILE

Missing Values

Critical

Columns Checked:	4
Columns with Issues:	3
Total Issues Found:	3
Affected Columns:	City, Phone, Sales

Business Impact

May produce inaccurate analytics and unreliable reporting.

Recommendation

Improve data collection at source; consider a required-field rule for critical columns.

Sample Findings

- City: 1 issue(s)
- Sales: 1 issue(s)
- Phone: 1 issue(s)

Duplicates

None

Columns Checked:	1
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Can create duplicate customers/orders and inflate revenue or count metrics.

Recommendation

Add a uniqueness constraint on the business key at the source system.

Type Mismatch

None

Columns Checked:	4
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Charting, grouping, and trend analysis may silently break or mislead.

Recommendation

Standardize the column's expected type at data entry.

Outliers

Critical

Columns Checked:	2
Columns with Issues:	1
Total Issues Found:	2
Affected Columns:	Sales

Business Impact

May distort averages, totals, and any ML model trained on this data.

Recommendation

Manually review flagged values before using this column in analysis or ML.

Sample Findings

- Sales: 2 issue(s)

Schema Quality

None

Columns Checked:	4
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Downstream tools and analysts may misread or skip unclear columns.

Recommendation

Rename unclear/duplicate/blank columns before this file is reused.

Consistency

High

Columns Checked:	4
Columns with Issues:	1
Total Issues Found:	1
Affected Columns:	City

Business Impact

Grouping and aggregation become unreliable (e.g. 'Paid' vs 'paid' split apart).

Recommendation

Standardize values at entry (dropdowns/lookups instead of free text).

Sample Findings

- City: 1 issue(s) examples=[{'normalized': 'lahore', 'canonical': 'Lahore', 'variants': {'Lahore': 1, 'lahore': 1}}]

Validity

High

Columns Checked:	5
Columns with Issues:	1
Total Issues Found:	2
Affected Columns:	Date

Business Impact

Values that look present may still be unusable or logically wrong.

Recommendation

Fix values that fail format/logic rules at the source before further use.

Sample Findings

- Date: 2 issue(s) rule=invalid_or_implausible_date

Freshness

High

Columns Checked:	4
Columns with Issues:	1
Total Issues Found:	1
Affected Columns:	Date

Business Impact

Decisions may be based on stale data without anyone noticing.

Recommendation

Confirm this data is refreshed on the expected schedule.

Sample Findings

- Date: 1 issue(s)

Column Quality Matrix

Column	Role	Score	Issues	Severity	Recommendation
City	identifier	67	2	High	Standardize values at entry (dropdowns/lookups instead
Sales	measurement	67	3	High	Improve data collection at source; consider a required-
Date	date	67	3	High	Confirm this data is refreshed on the expected schedule
Phone	pii	83	1	Medium	Improve data collection at source; consider a required-

Top Critical Issues

1. **Missing Values -- City**

   Critical

   Impact: May produce inaccurate analytics and unreliable reporting.
   Action: Improve data collection at source; consider a required-field rule for critical columns.

2. **Missing Values -- Phone**

   Critical

   Impact: May produce inaccurate analytics and unreliable reporting.
   Action: Improve data collection at source; consider a required-field rule for critical columns.

3. **Missing Values -- Sales**

   Critical

   Impact: May produce inaccurate analytics and unreliable reporting.
   Action: Improve data collection at source; consider a required-field rule for critical columns.

4. **Outliers -- Sales**

   Critical

   Impact: May distort averages, totals, and any ML model trained on this data.
   Action: Manually review flagged values before using this column in analysis or ML.

5. **Consistency -- City**

   High

   Impact: Grouping and aggregation become unreliable (e.g. 'Paid' vs 'paid' split apart).
   Action: Standardize values at entry (dropdowns/lookups instead of free text).

6. **Validity -- Date**

   High

   Impact: Values that look present may still be unusable or logically wrong.
   Action: Fix values that fail format/logic rules at the source before further use.

7. **Freshness -- Date**

   High

   Impact: Decisions may be based on stale data without anyone noticing.
   Action: Confirm this data is refreshed on the expected schedule.

Appendix

Execution Time: 0.05s

Engine Version: Phase 1

Scoring Formula

Each dimension score = 100 x (passed checks / total non-error checks) for that dimension. Composite score = weighted average across scorable dimensions only (weights re-normalized when a dimension has no results). Privacy Risk is calculated and reported separately and is never subtracted from the composite score.

Quality Dimensions

- Completeness (weight: 20%)
- Validity (weight: 20%)
- Type Reliability (weight: 15%)
- Consistency (weight: 15%)
- Uniqueness (weight: 10%)
- Schema Quality (weight: 10%)
- Outlier Risk (weight: 5%)
- Freshness (weight: 5%)